Chapter 1: Probability Theory Foundations

Part I — Statistics Foundation for ML

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Abstract

Before a single model gets fit, every statistical and machine-learning method rests on the same small set of ideas: a sample space, an assignment of probabilities to events, and the machinery – random variables, distribution functions, densities – that lets us reason about randomness in a form we can compute with. This chapter builds that foundation from first principles and ties each idea back to a machine-learning example, so that later chapters (regression, classification, EM, attention weights as probability distributions, and so on) can simply say "recall Chapter 1" instead of re-deriving the axioms of probability from scratch.

Acknowledgement: this chapter expands the STAT4ML lecture slides ("Slide 1 – Probability Theory Foundation"), which were themselves prepared following Casella & Berger, Statistical Inference*, 2nd ed. (2002).*

Contents

1 Sample Space

Definition 1.1 (Sample space)**.** The set of all possible outcomes of a particular experiment is called the *sample space* of the experiment, generally denoted Ω.

Example 1.1 (Two coins)**.** Tossing two fair coins has sample space

$$\Omega = \{HH, HT, TH, TT\}.$$

The notion of "experiment" is deliberately abstract: it can be a physical coin toss, but it can just as well be "a user visits a web page" or "a model receives one input and produces one output." Making the sample space explicit is the first step in turning an ML problem into a probability problem.

Try it yourself

- What is the sample space of rolling a fair six-sided die?
- What is the sample space of a credit-risk decision (default / no default within 12 months)?
- What is the sample space of a binary classification problem with labels $\{0, 1\}$?
- Define the sample space of a single visit to a specific web page (think of every clickable element as a possible outcome).
- Define the sample space of one turn of a chatbot conversation.

1.1 Events

Definition 1.2 (Event space)**.** The *event space* $\mathcal{A}$ is the collection of potential results of the experiment we are willing to assign a probability to. In the discrete case, $\mathcal{A}$ is the power set of Ω.

Definition 1.3 (Event)**.** An *event* is any collection of possible outcomes of an experiment – that is, any subset of Ω. The triple $(\Omega, \mathcal{A}, P)$ is called a *probability space*.

Example 1.2. On an Amazon product page, the whole page is the sample space and every clickable item – "Add to Cart", "See Similar Items", etc. – is an event.

2 Probability Theory Foundations

2.1 Axiomatic Foundations

Definition 2.1 (σ-algebra)**.** A collection $\mathcal{B}$ of subsets of Ω is a *sigma algebra* (σ-algebra) if it satisfies:

1. $\emptyset \in \mathcal{B}$.
2. If $A \in \mathcal{B}$, then $A^c \in \mathcal{B}$ ($\mathcal{B}$ is closed under complement).
3. If $A_1, A_2, \cdots \in \mathcal{B}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$ ($\mathcal{B}$ is closed under countable unions).

The pair $(\Omega, \mathcal{B})$ is called a *measurable space* (or Borel space). Interpretation: a σ-algebra is exactly the collection of events we are allowed to assign probabilities to.

A short derivation worth doing once by hand: closure under countable *intersections* follows from properties (2) and (3) together with De Morgan's law. If $A_1, A_2, \cdots \in \mathcal{B}$ then $A_1^c, A_2^c, \cdots \in \mathcal{B}$ by (2); by (3), $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{B}$; and De Morgan's law gives

$$\Big(\bigcup_{i=1}^{\infty} A_i^c\Big)^c = \bigcap_{i=1}^{\infty} A_i,$$

which is in $\mathcal{B}$ by (2) again. So a σ-algebra is closed under countable intersections as well as countable unions.

Example 2.1. If Ω is finite or countable, the simplest choice is $\mathcal{B}$ = (all subsets of Ω, including Ω itself), i.e. the power set of Ω. For $\Omega = \{A, B, \ldots, Z\}$, $\mathcal{B}$ is the power set of Ω, with 2^{26} elements.

Try it yourself

- Is the power set of Ω itself countable, when Ω is countably infinite? (No – this is Cantor's theorem.)
- Define an event space for a single web page. Check that it satisfies all three σ-algebra properties.

2.2 Kolmogorov's Axioms

Definition 2.2 (Probability function)**.** Given a sample space Ω and an associated σ-algebra $\mathcal{B}$, a *probability function* is a function P with domain $\mathcal{B}$ satisfying:

1. $P(A) \geq 0$ for all $A \in \mathcal{B}$.
2. $P(\Omega) = 1$.
3. If $A_1, A_2, \cdots \in \mathcal{B}$ are pairwise disjoint, then $P\big(\bigcup_{i=1}^{\infty} A_i\big) = \sum_{i=1}^{\infty} P(A_i)$ (countable additivity).

These are the *axioms of probability*, also known as *Kolmogorov's axioms*. Any function $P(\cdot)$ satisfying them is a valid probability function – there is no other requirement.

Example 2.2 (Fair coin)**.** Toss a fair coin once: $\Omega = \{H, T\}$. Since $\Omega = \{H\} \cup \{T\}$ and the two outcomes are disjoint, Axiom 3 gives $P(\{H\} \cup \{T\}) = P(\{H\}) + P(\{T\})$, and Axiom 2 gives $P(\{H\} \cup \{T\}) = 1$. So $P(\{H\}) + P(\{T\}) = 1$. The axioms alone do *not* force $P(\{H\}) = P(\{T\}) = 1/2$ – any nonnegative pair summing to 1 (e.g. $P(\{H\}) = 1/4$, $P(\{T\}) = 3/4$) is axiomatically valid. We choose $1/2$ each because we additionally know, empirically, that the coin is fair. *The axioms tell you what a probability function is allowed to be; they never tell you which one describes your data. That second step is statistics.*

Theorem 2.1 (Elementary consequences)**.** If P is a probability function and $A, B \in \mathcal{B}$, then

1. $P(\emptyset) = 0$.
2. $P(A) \leq 1$.
3. $P(A^c) = 1 - P(A)$.
4. $P(B \cap A^c) = P(B) - P(A \cap B)$.
5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (*inclusion–exclusion*).
6. If $A \subset B$, then $P(A) \leq P(B)$.

Property 5 above is the one most often mis-remembered as $P(A \cup B) = P(A) + P(B)$ – correct only when A and B are disjoint. It reappears, unfixed, as a classic bug in the Python/R/Rust exercises for this chapter.

3 Independence

Definition 3.1. Two events A and B are *statistically independent* if

$$P(A \cap B) = P(A)\,P(B).$$

Theorem 3.1. If A and B are independent, then so are each of the pairs (A, B^c), (A^c, B), and (A^c, B^c).

4 Conditional Probability

Definition 4.1. If A and B are events with $P(B) > 0$, the *conditional probability* of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \tag{1}$$

In words: given that B has occurred, what is the probability that A also occurred? By definition $P(B \mid B) = 1$, and if A, B are disjoint then $P(A \mid B) = P(B \mid A) = 0$. Conditioning on B effectively shrinks the sample space to B; it is a short exercise to check that $P(\cdot \mid B)$ again satisfies Kolmogorov's axioms for any fixed B with $P(B) > 0$.

4.1 An NLP Illustration

A famous distributional-semantics slogan is: "you shall know a word by the company it keeps" (Firth, 1957). Word2Vec-style models turn this into a conditional-probability estimation problem: for a center word w_t at position t, and a context window of size m, estimate

$$P(w_{t+i} \mid w_t), \qquad i \in \{-m, \ldots, -1, 1, \ldots, m\}.$$

This is exactly Equation (**??**) applied at corpus scale: every skip-gram training pair is one empirical conditional-probability query.

4.2 Bayes' Rule and Bayes' Theorem

Since $P(A \mid B) = P(A \cap B)/P(B)$ and $P(B \mid A) = P(A \cap B)/P(A)$,

$$P(A \mid B)\, P(B) = P(A \cap B) = P(B \mid A)\, P(A) \quad \Longrightarrow \quad P(A \mid B) = P(B \mid A)\, \frac{P(A)}{P(B)}.$$

This is *Bayes' rule*.

Theorem 4.1 (Bayes' theorem)**.** Let $A_1, A_2, \ldots$ be a partition of the sample space, and let B be any event. Then for each $i = 1, 2, \ldots$

$$P(A_i \mid B) = \frac{P(B \mid A_i)\, P(A_i)}{\sum_{j=1}^{\infty} P(B \mid A_j)\, P(A_j)}.$$

The denominator is just $P(B)$ rewritten via the law of total probability: $\sum_j P(B \mid A_j) P(A_j) = \sum_j P(A_j \cap B) = P(B)$, using the fact that $A_1, A_2, \ldots$ partition Ω.

5 Random Variables

Definition 5.1. A *random variable* is a function from the sample space Ω into the real numbers $\mathbb{R}$.

Experiment	Random variable
Click on an Amazon product page	$X = \mathbb{1}$[clicked BUY button]
Product-page visit	$X =$ total GET-API calls to the page

Example 5.1. Toss a fair coin three times and let X be the number of heads:

ω	HHH	HHT	HTH	THH	TTH	THT	HTT	TTT
$X(\omega)$	3	2	2	2	1	1	1	0

Formally, if $X : \Omega \to \mathcal{X}$ maps into an induced sample space $\mathcal{X} = \{x_1, \dots, x_m\}$, the *induced probability function* P_X is defined by

$$P_X(X = x_i) = P(\{\omega \in \Omega : X(\omega) = x_i\}),$$

and for uncountable $\mathcal{X}$, by $P_X(X \in A) = P(\{\omega : X(\omega) \in A\})$ for $A \subset \mathcal{X}$. P_X satisfies Kolmogorov's axioms because P does – this is worth checking directly at least once.

Convention: capital letters (X) denote random variables, lower-case letters (x) denote the particular value they take.

6 Probability Function

6.1 Distribution Functions

Definition 6.1 (CDF)**.** The *cumulative distribution function* of a random variable X, $F_X(x)$, is

$$F_X(x) = P_X(X \le x), \qquad \text{for all } x \in \mathbb{R}.$$

Example 6.1. Toss three fair coins and let X be the number of heads. Then

$$F_X(x) = \begin{cases} 0 & -\infty < x < 0 \\ 1/8 & 0 \le x < 1 \\ 1/2 & 1 \le x < 2 \\ 7/8 & 2 \le x < 3 \\ 1 & 3 \le x < \infty \end{cases}$$

so, for instance, $P(X \le 2.5) = P(X \in \{0, 1, 2\}) = 7/8$.

Theorem 6.1. F_X is a CDF if and only if

1. $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to +\infty} F_X(x) = 1$,
2. F_X is non-decreasing,
3. F_X is right-continuous: for every x_0, $\lim_{x \downarrow x_0} F_X(x) = F_X(x_0)$.

X is *continuous* (resp. *discrete*) if F_X is a continuous (resp. step) function of x.

Example 6.2 (The logistic sigmoid as a CDF)**.** Take $F_X(x) = \dfrac{1}{1 + e^{-x}}$, the logistic sigmoid familiar from every neural-network output layer. Then $\lim_{x \to -\infty} F_X(x) = 0$, $\lim_{x \to \infty} F_X(x) = 1$, and

$$\frac{d}{dx} F_X(x) = \frac{e^{-x}}{(1 + e^{-x})^2} > 0,$$

so F_X is non-decreasing and continuous everywhere – it *is* a valid CDF (of the standard logistic distribution), which is exactly why it is a sensible choice of squashing function for turning a real-valued logit into something we can read off as a probability.

Theorem 6.2. X and Y are identically distributed if and only if $F_X(x) = F_Y(x)$ for every x.

6.2 Density Function

Definition 6.2 (pmf)**.** For a discrete random variable, the *probability mass function* is $f_X(x) = P(X = x)$ for all x.

Definition 6.3 (pdf)**.** For a continuous random variable, the *probability density function* is the function $f_X(x)$ satisfying

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

We write "$X \sim F_X$" or "$X \sim f_X$" to mean "X has distribution F_X (or density f_X)"; if X and Y share a distribution we write $X \sim Y$.

Example 6.3. For the logistic CDF above, $f_X(x) = \dfrac{e^{-x}}{(1+e^{-x})^2}$, and for continuous X in general,

$$P(a < X < b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\,dx.$$

Theorem 6.3. f_X is a valid pdf/pmf of some random variable X if and only if $f_X(x) \geq 0$ for all x, and $\sum_x f_X(x) = 1$ (discrete) or $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ (continuous).

This gives a useful recipe for *simulation*: if $h(x) \geq 0$ on some set A and $0 \leq \int_A h(x)\,dx = K < \infty$, then $f_X(x) = h(x)/K$ is a valid density on A. (Note that a CDF can exist and be continuous without being differentiable everywhere, in which case f_X does not exist and analysis has to proceed via F_X directly.)

7 Distributions of Functions of a Random Variable

If X has CDF F_X and $Y = g(X)$, then Y is also a random variable, and for any set A,

$$P(Y \in A) = P(g(X) \in A) = P(X \in g^{-1}(A)), \qquad g^{-1}(A) = \{x \in \mathcal{X} : g(x) \in A\}.$$

Consequently the CDF of Y is

$$F_Y(y) = P(g(X) \leq y) = \int_{x:g(x)\leq y} f_X(x)\,dx.$$

Theorem 7.1. Let X have CDF F_X, $Y = g(X)$, $\mathcal{X} = \{x : f_X(x) > 0\}$, $\mathcal{Y} = \{y : y = g(x), x \in \mathcal{X}\}$.

1. If g is increasing on $\mathcal{X}$, then $F_Y(y) = F_X(g^{-1}(y))$ for $y \in \mathcal{Y}$.
2. If g is decreasing on $\mathcal{X}$ and X is continuous, then $F_Y(y) = 1 - F_X(g^{-1}(y))$ for $y \in \mathcal{Y}$.

Theorem 7.2 (Change of variables for densities)**.** Let X have pdf f_X, $Y = g(X)$ with g monotone, f_X continuous on $\mathcal{X}$, and g^{-1} continuously differentiable on $\mathcal{Y}$. Then

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \dfrac{d}{dy} g^{-1}(y) \right| & y \in \mathcal{Y} \\ 0 & \text{otherwise.} \end{cases}$$

Example 7.1. Let $X \sim \text{Uniform}(0,1)$, so $f_X(x) = 1$ on $(0,1)$ and $F_X(x) = x$. Let $Y = g(X) = -\log X$. Then $g'(x) = -1/x < 0$ on $(0,1)$, so g is monotone with a continuous derivative, $\mathcal{Y} = (0,\infty)$, and $g^{-1}(y) = e^{-y}$. By the change-of-variables theorem,

$$f_Y(y) = 1 \cdot |-e^{-y}| = e^{-y}, \qquad y > 0,$$

i.e. $Y \sim \text{Exponential}(1)$. This is the standard trick behind *inverse-transform sampling*.

Theorem 7.3 (Probability integral transform)**.** Let X have continuous CDF F_X and define $Y = F_X(X)$. Then $Y \sim \text{Uniform}(0,1)$.

This result is the workhorse of Monte Carlo simulation: to sample from any distribution with CDF F_X, sample $Y \sim \text{Uniform}(0,1)$ and solve $F_X(x) = y$ for x, i.e. return $x = F_X^{-1}(y)$.

8 Summary

Summary

- A sample space Ω, a σ-algebra $\mathcal{B}$ of events, and a probability function P satisfying Kolmogorov's axioms together define a *probability space*.
- Conditional probability and Bayes' theorem let us update probabilities in light of evidence – the mathematical basis of every Bayesian model in this course, and of skip-gram word embeddings.
- A random variable maps outcomes to real numbers; its CDF and pdf/pmf fully describe its randomness.
- The change-of-variables formula and the probability integral transform connect arbitrary distributions to the uniform distribution – the basis of random-number generation.

9 Exercises

See `exercises/python/exercise01.py`, `exercises/r/exercise01.R`, and `exercises/rust/exercise01.rs` for a working implementation of a small "probability toolkit" (axiom checking, inclusion–exclusion, Bayes' theorem, and a Monte-Carlo law-of-large-numbers check), plus `exercise01_buggy.{py,R,rs}` a deliberately broken version of the same toolkit that you must repair function by function.

Reference

Casella, G., & Berger, R. (2002). *Statistical Inference* (2nd ed.). Cengage Learning.